# Practice workbook · Compression: lossy or lossless?

Why files are compressed, drawbacks of compression, the meaning and examples of lossy and lossless, and choosing and justifying a method for a scenario.

9 question parts: 1 worked-example retry and 8 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T3-14

# 1. Question 5 · part d · 3 marks

2024 JUN · Paper 01

Worked-example retry · Attempt part d only.

**5** A team is developing a computer game.

(a) Complete the table by adding a tick (✓) to show the characteristics of program language translators.

You may select more than one translator per characteristic.

**(3)**

| Characteristic | Translator | | |
|---|---|---|---|
| | **Assembler** | **Interpreter** | **Compiler** |
| Does **not** have to be present in memory to execute the translated code. | | | |
| Translates high level language source code. | | | |
| Executes high level code until either the program has finished, or a syntax error is encountered. | | | |
| Translates low level language source code into machine code. | | | |

(b) Luca is testing the computer game.

The computer that he is using is an older model with a processor that runs at a low clock speed.

Explain **one** drawback of testing the game using this computer.

**(2)**

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# 1. Question 5 · part d · 3 marks

2024 JUN · Paper 01

Worked-example retry · Attempt part d only.

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(c) Anna uses utility software.

Complete the table by adding the type of utility software she should use to solve each issue.

| Issue | Utility software |
|---|---|
| Her word processor cannot read all of a file because it is corrupt. | |
| She has a JPEG image and needs to send it to a publisher who can only read PNG image files. | |
| There has been a flood and files on a server have been damaged, but Anna needs to access her data. | |
| Her hard drive is running very slowly when she tries to access a file. | |

(4)

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# 1. Question 5 · part d · 3 marks

2024 JUN · Paper 01

Worked-example retry · Attempt part d only.

(d) Tarik is writing an instruction manual for the computer game.

He decides to use software to compress the instruction manual before emailing it to a proofreader for checking.

State and justify the most appropriate compression method.

(3)

Method

.....

Justification

.....

.....

.....

.....

(e) Members of the team developing the game are either coders or designers.

Coders (C) write the application code.

Designers (D) design the graphics for the game and write instruction manuals.

Team members have one role only.

Construct a logic statement, using AND, OR and NOT with the letters C and D, to show the conditions for being a member of the team.

(3)

**(Total for Question 5 = 15 marks)**

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## 1. Question 5 · part d · 3 marks · Working and answer

Working and answer

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# 2. Question 4 · part c.i · 2 marks

2019 JUN · Paper 01

Further practice · Attempt part c.i only.

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**4** One function of an operating system is file management.

(a) Both kibibyte and kilobyte can be used as measures of file size.

Compare kibibyte and kilobyte.

**(2)**

(b) An image is 2322 pixels high and 4128 pixels wide.

The image is stored with a 16-bit colour depth.

The metadata for the image is 975 bytes.

Construct an expression to show how the file size, in megabytes, is calculated.

You do **not** need to do the calculation.

**(4)**

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## 2. Question 4 · part c.i · 2 marks

2019 JUN · Paper 01

Further practice · Attempt part c.i only.

(c) Operating systems often include compression software for reducing file sizes.

(i) Give **two** reasons for reducing file sizes.

(2)

1 .....

.....

2 .....

.....

(ii) Give **two** drawbacks of using compression software.

(2)

1 .....

.....

2 .....

.....

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## 2. Question 4 · part c.i · 2 marks · Working and answer

Working and answer

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# 3. Question 4 · part c.ii · 2 marks

2019 JUN · Paper 01

Further practice · Attempt part c.ii only.

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

**4** One function of an operating system is file management.

(a) Both kibibyte and kilobyte can be used as measures of file size.

Compare kibibyte and kilobyte.

**(2)**

(b) An image is 2322 pixels high and 4128 pixels wide.

The image is stored with a 16-bit colour depth.

The metadata for the image is 975 bytes.

Construct an expression to show how the file size, in megabytes, is calculated.

You do **not** need to do the calculation.

**(4)**

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## 3. Question 4 · part c.ii · 2 marks

2019 JUN · Paper 01

Further practice · Attempt part c.ii only.

(c) Operating systems often include compression software for reducing file sizes.

(i) Give **two** reasons for reducing file sizes.

(2)

1 .....

.....

2 .....

.....

(ii) Give **two** drawbacks of using compression software.

(2)

1 .....

.....

2 .....

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## 3. Question 4 · part c.ii · 2 marks

2019 JUN · Paper 01

Further practice · Attempt part c.ii only.

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(d) A text file is stored on a hard disc.

The file holds information from one side of a sheet of paper.

The sheet of paper is represented as a grid, 80 columns wide and 66 rows long.

Each cell in the grid contains a single 2-byte Unicode character.

The file also contains 40 characters of metadata.

The hard disc allocates space in blocks of 1024 bytes.

Construct an expression to show the number of blocks required to store the file.

You do **not** need to do the calculation.

**(4)**

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**(Total for Question 4 = 14 marks)**

## 3. Question 4 · part c.ii · 2 marks · Working and answer

Working and answer

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# 4. Question 4 · part c · 1 mark

2020 NOV · Paper 01

Further practice · Attempt part c only.

**4** Ibrahim works in the music industry.

(a) He is learning how sound is represented in binary.

**Figure 3** illustrates this process.

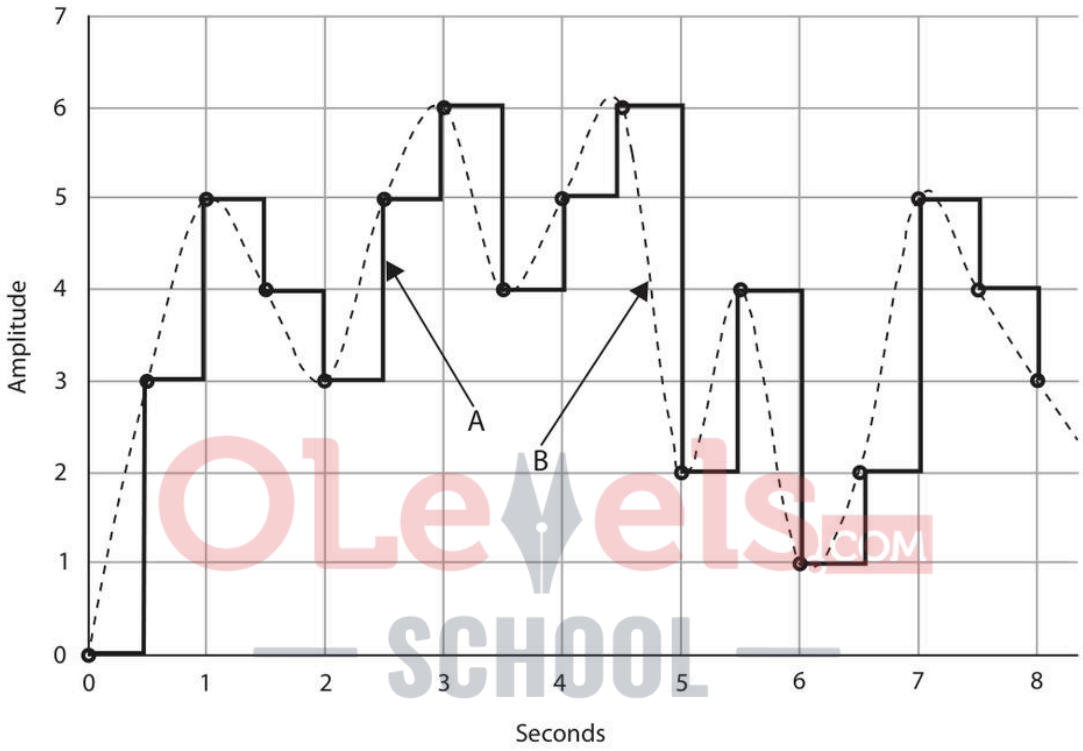



**Figure 3**

(i) State what the solid line labelled **A** represents.

**(1)**

(ii) State what the dashed line labelled **B** represents.

**(1)**

(iii) Give the sampling frequency, including the correct units.

**(1)**

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DO NOT WRITE IN THIS AREA

**10**

P 6 1 8 8 4 R R A 0 1 0 2 0

# 4. Question 4 · part c · 1 mark

2020 NOV · Paper 01

Further practice · Attempt part c only.

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DO NOT WRITE IN THIS AREA

(iv) Give **one** impact of increasing the sampling frequency.

(1)

(v) The bit-depth measures the resolution of the sound sample.

State the minimum bit-depth needed to store the sound sampled in **Figure 3**.

Justify your answer.

(2)

Bit-depth

Justification

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(vi) Give the value in binary of the audio sample at the 4th second.

(1)

# 4. Question 4 · part c · 1 mark

2020 NOV · Paper 01

Further practice · Attempt part c only.

(b) Ibrahim wants to store a music file.

- The sampling frequency is 47 kHz.
- The bit-depth is 64
- The analogue signal lasts for 819 seconds.

Construct an expression to show how the file size, in bytes, is calculated.

You do **not** need to do the calculation.

**(4)**

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(c) Ibrahim wants to upload a music file to his online storage. The file has been compressed using a lossy algorithm.

Give **one** drawback of using a lossy algorithm for this purpose.

**(1)**

**(Total for Question 4 = 12 marks)**

## 4. Question 4 · part c · 1 mark · Working and answer

Working and answer

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# 5. Question 2 · part f.i · 2 marks

2021 JUN · Paper 01

Further practice · Attempt part f.i only.

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**2** Many organisations use networks.

(a) One reason for networking computers is to share access to the internet.

Give **one other** reason for connecting computers in a local area network.

**(1)**

(b) State the purpose of network protocols.

**(1)**

(c) Data is transmitted in packets.

(i) Identify the layer of the TCP/IP model that would add the File Transfer Protocol (FTP).

**(1)**

- ☒ **A** Application
- ☒ **B** Data
- ☒ **C** Network
- ☒ **D** Transport

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(ii) Give **one** task carried out by the network layer.

**(1)**

(d) Complete the table by adding a tick (✓) to match each item to its name.

**(3)**

| Item | Domain Name | IP Address | URL |
|---|---|---|---|
| https://www.pearson.com | | | |
| 216.33.251.195 | | | |
| pearson.com | | | |

# 5. Question 2 · part f.i · 2 marks

2021 JUN · Paper 01

Further practice · Attempt part f.i only.

(e) **Figure 2** shows a network topology.

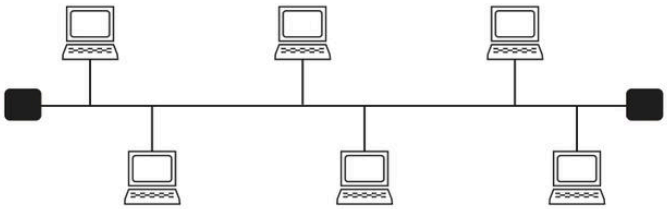


**Figure 2**

(i) State the name of this network topology.

**(1)**

(ii) State **one** benefit of using this network topology.

**(1)**

(iii) State **one** drawback of using this network topology.

**(1)**

(f) Files may be compressed using lossless or lossy algorithms.

(i) Music files are distributed over the internet.

Give **two** reasons why lossy compression is used when distributing music files over the internet.

**(2)**

1

2

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## 5. Question 2 · part f.i · 2 marks

2021 JUN · Paper 01

Further practice · Attempt part f.i only.

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(ii) Identify which **one** of these file formats uses lossy compression.

(1)

- ☒ **A** BMP
- ☒ **B** DOC
- ☒ **C** JPEG
- ☒ **D** PNG

(iii) Here is a string of data.

**CCCWWWCCWWWWWWCCC**

Give the result of compressing the string using a run-length encoding algorithm.

(1)

DO NOT WRITE IN THIS AREA

(g) A video file is to be transmitted over the internet.

- The network transmission speed is 54 Mbps.
- The file size is 6 gigabytes (GB).

Construct an expression to show how the transmission time, in seconds, is calculated.

You do **not** have to do the calculation.

(4)

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**(Total for Question 2 = 18 marks)**

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## 5. Question 2 · part f.i · 2 marks · Working and answer

Working and answer

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# 6. Question 2 · part f.ii · 1 mark

2021 JUN · Paper 01

Further practice · Attempt part f.ii only.

DO NOT WRITE IN THIS AREA

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DO NOT WRITE IN THIS AREA

**2** Many organisations use networks.

(a) One reason for networking computers is to share access to the internet.

Give **one other** reason for connecting computers in a local area network.

**(1)**

(b) State the purpose of network protocols.

**(1)**

(c) Data is transmitted in packets.

(i) Identify the layer of the TCP/IP model that would add the File Transfer Protocol (FTP).

**(1)**

- ☒ **A** Application
- ☒ **B** Data
- ☒ **C** Network
- ☒ **D** Transport

(ii) Give **one** task carried out by the network layer.

**(1)**

(d) Complete the table by adding a tick (✓) to match each item to its name.

**(3)**

| Item | Domain Name | IP Address | URL |
|---|---|---|---|
| https://www.pearson.com | | | |
| 216.33.251.195 | | | |
| pearson.com | | | |

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# 6. Question 2 · part f.ii · 1 mark

2021 JUN · Paper 01

Further practice · Attempt part f.ii only.

(e) **Figure 2** shows a network topology.

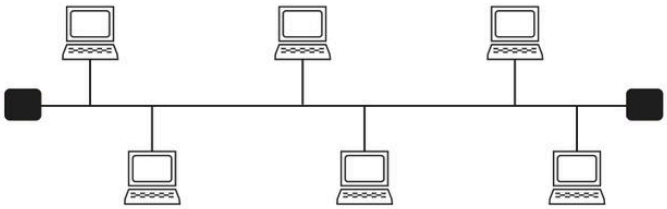


**Figure 2**

(i) State the name of this network topology.

**(1)**

(ii) State **one** benefit of using this network topology.

**(1)**

(iii) State **one** drawback of using this network topology.

**(1)**

(f) Files may be compressed using lossless or lossy algorithms.

(i) Music files are distributed over the internet.

Give **two** reasons why lossy compression is used when distributing music files over the internet.

**(2)**

1 .....

2 .....

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6

P 6 6 4 8 9 A 0 6 2 0

# 6. Question 2 · part f.ii · 1 mark

2021 JUN · Paper 01

Further practice · Attempt part f.ii only.

DO NOT WRITE IN THIS AREA

(ii) Identify which **one** of these file formats uses lossy compression.

(1)

- ☒ **A** BMP
- ☒ **B** DOC
- ☒ **C** JPEG
- ☒ **D** PNG

(iii) Here is a string of data.

**CCCWWWCCWWWWWWCCC**

Give the result of compressing the string using a run-length encoding algorithm.

(1)

.....

.....

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(g) A video file is to be transmitted over the internet.

- The network transmission speed is 54 Mbps.
- The file size is 6 gigabytes (GB).

Construct an expression to show how the transmission time, in seconds, is calculated.

You do **not** have to do the calculation.

(4)

DO NOT WRITE IN THIS AREA

**(Total for Question 2 = 18 marks)**

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## 6. Question 2 · part f.ii · 1 mark · Working and answer

Working and answer

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## 7. Question 1 · part b.iii · 1 mark

2021 NOV · Paper 01

Further practice · Attempt part b.iii only.

**1** Computers use binary to represent and store data.

(a) Binary and hexadecimal notation can be used to represent numbers.

(i) Convert the denary number 77 to 8-bit binary.

**(2)**

(ii) Convert the denary number –126 to 8-bit binary using two's complement.

**(2)**

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(iii) Convert the binary pattern 11000110 to hexadecimal.

**(2)**

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## 7. Question 1 · part b.iii · 1 mark

2021 NOV · Paper 01

Further practice · Attempt part b.iii only.

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(b) Images are stored as binary data.

(i) A bitmap image is 400 pixels wide by 200 pixels high.

It has a colour depth of 12 bits.

Construct an expression to show how the image size, in bytes, is calculated.

Do **not** calculate the answer.

**(2)**

(ii) **Figure 1** is an image of an icon, set out in an 8 × 8 pixel grid.

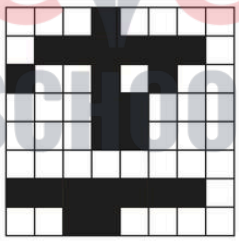


**Figure 1**

A run-length encoding compression algorithm uses:

- W for white pixels
- B for black pixels.

Give the result of compressing the first 4 lines of the icon using the algorithm.

**(3)**

Line 1 .....

Line 2 .....

Line 3 .....

Line 4 .....

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# 7. Question 1 · part b.iii · 1 mark

2021 NOV · Paper 01

Further practice · Attempt part b.iii only.

(iii) Run-length encoding is a type of lossless compression.

State what is meant by the term **lossless** in this context.

**(1)**

(c) Raoul wants to record a song and store it as a digital sound file.

**Figure 2** shows an overview of the process of recording a song and storing it as a digital sound file.

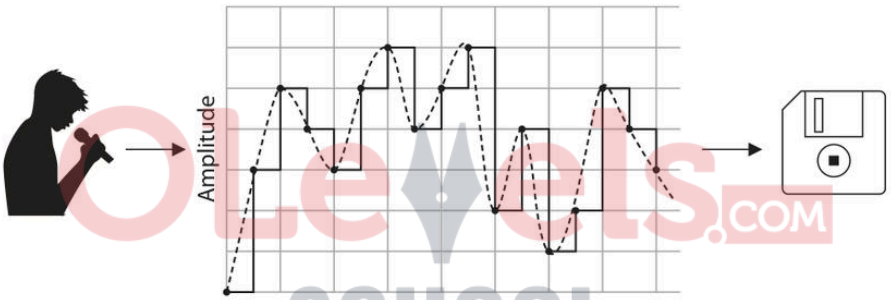



**Figure 2**

(i) Describe the steps taken to convert the analogue sound to a digital sound file.

**(3)**

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4

P 6 9 2 9 7 A 0 4 2 0

## 7. Question 1 · part b.iii · 1 mark · Working and answer

Working and answer

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## 8. Question 1 · part c.ii · 1 mark

2021 NOV · Paper 01

Further practice · Attempt part c.ii only.

**1** Computers use binary to represent and store data.

(a) Binary and hexadecimal notation can be used to represent numbers.

(i) Convert the denary number 77 to 8-bit binary.

**(2)**

(ii) Convert the denary number –126 to 8-bit binary using two's complement.

**(2)**

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(iii) Convert the binary pattern 11000110 to hexadecimal.

**(2)**

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## 8. Question 1 · part c.ii · 1 mark

2021 NOV · Paper 01

Further practice · Attempt part c.ii only.

(b) Images are stored as binary data.

(i) A bitmap image is 400 pixels wide by 200 pixels high.

It has a colour depth of 12 bits.

Construct an expression to show how the image size, in bytes, is calculated.

Do **not** calculate the answer.

**(2)**

(ii) **Figure 1** is an image of an icon, set out in an 8 × 8 pixel grid.

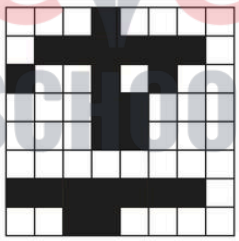


**Figure 1**

A run-length encoding compression algorithm uses:

- W for white pixels
- B for black pixels.

Give the result of compressing the first 4 lines of the icon using the algorithm.

**(3)**

Line 1 .....

Line 2 .....

Line 3 .....

Line 4 .....

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# 8. Question 1 · part c.ii · 1 mark

2021 NOV · Paper 01

Further practice · Attempt part c.ii only.

(iii) Run-length encoding is a type of lossless compression.

State what is meant by the term **lossless** in this context.

**(1)**

(c) Raoul wants to record a song and store it as a digital sound file.

**Figure 2** shows an overview of the process of recording a song and storing it as a digital sound file.

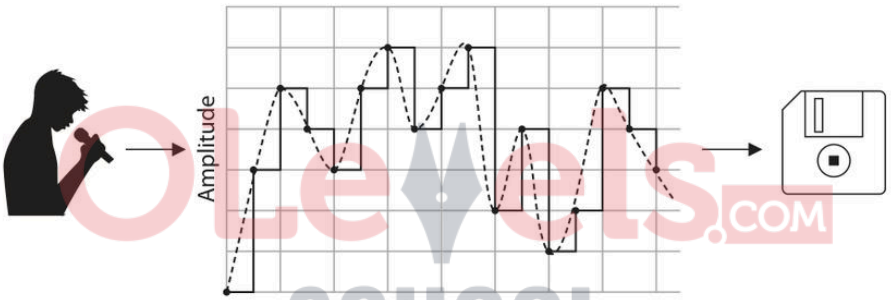



**Figure 2**

(i) Describe the steps taken to convert the analogue sound to a digital sound file.

**(3)**

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DO NOT WRITE IN THIS AREA

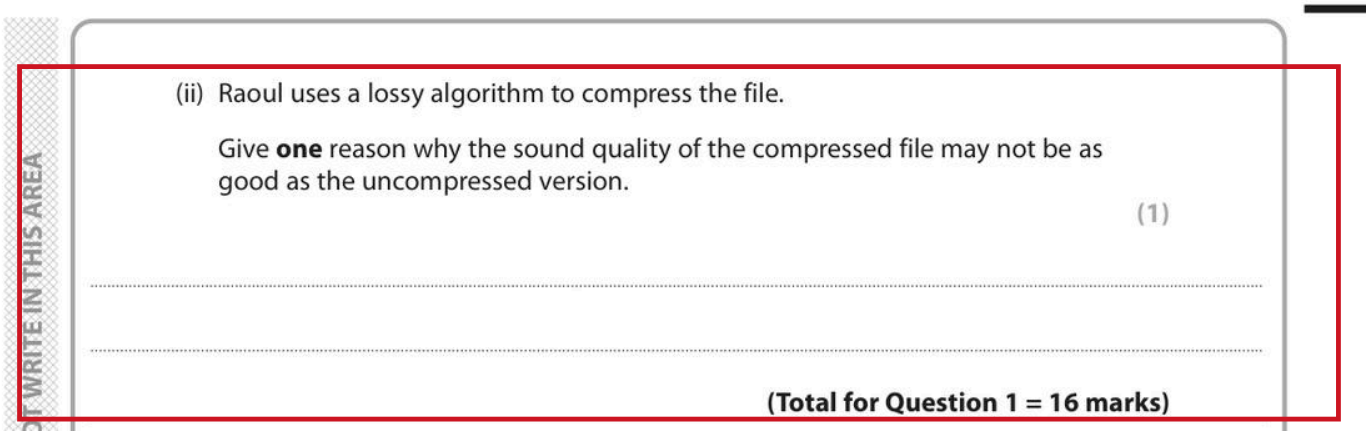


P 6 9 2 9 7 A 0 4 2 0

## 8. Question 1 · part c.ii · 1 mark

2021 NOV · Paper 01

Further practice · Attempt part c.ii only.



(ii) Raoul uses a lossy algorithm to compress the file.

Give **one** reason why the sound quality of the compressed file may not be as good as the uncompressed version.

**(1)**

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**(Total for Question 1 = 16 marks)**

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Working and answer

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# 9. Question 3 · part b.i · 1 mark

2022 JUN · Paper 01

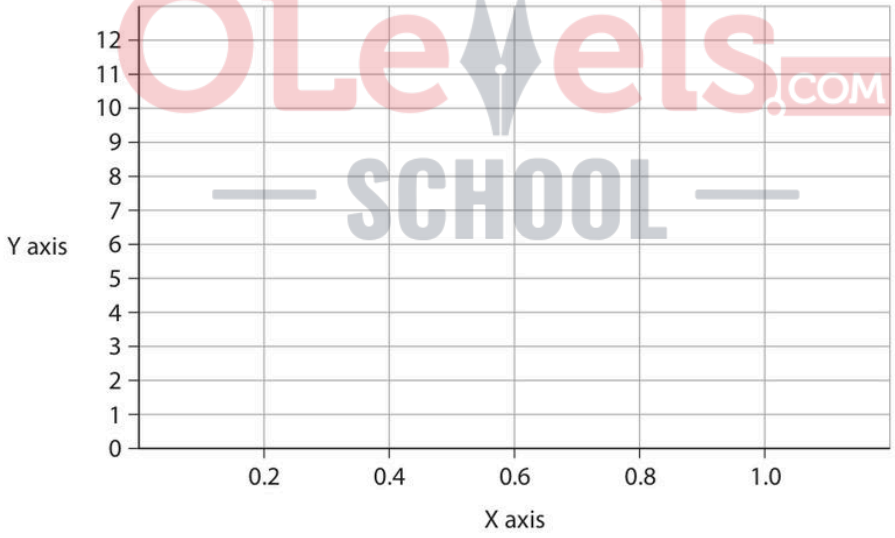
Further practice · Attempt part b.i only.

**3** Alyssa is a music producer.

(a) **Figure 1** shows the denary values of five samples of an analogue sound using a sample interval of 0.2 seconds.

| Sample number | Denary value |
|---|---|
| 1 | 1 |
| 2 | 10 |
| 3 | 12 |
| 4 | 5 |
| 5 | 3 |

**Figure 1**




(i) Complete this graph using the sample information from **Figure 1** to show the digital sound wave.

**(3)**

(ii) Give a suitable label for the X axis.

**(1)**

(iii) Give a suitable label for the Y axis.

**(1)**

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8

P 7 2 5 3 8 A 0 8 2 0

## 9. Question 3 · part b.i · 1 mark

2022 JUN · Paper 01

Further practice · Attempt part b.i only.

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(b) Alyssa uploads music files to her cloud storage.

(i) She compresses the files before she uploads them using a lossless algorithm.

Give **one** disadvantage of using a lossless rather than a lossy algorithm for this purpose.

**(1)**

.....

.....

(ii) Explain **one** benefit to Alyssa of storing her music files in the cloud.

**(2)**

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.....

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(iii) Give **one** possible security issue associated with storing music files in the cloud.

**(1)**

.....

.....

(iv) One of Alyssa's music files is stored at https://www.cloudisfab.com/re12/ru2.mp3

Complete the table by adding a description of each URL component.

**(4)**

| URL component | Description |
|---|---|
| https | |
| www.cloudisfab.com | |
| re12 | |
| ru2.mp3 | |

**(Total for Question 3 = 13 marks)**

## 9. Question 3 · part b.i · 1 mark · Working and answer

Working and answer

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